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$$f(x) = x + \frac{1}{x} \left(= \frac{x^2+1}{x} \right) \to \text{nulw: } x = 0$$

VA:

$$\lim_{x\to 0} \frac{x^2+1}{x} = \left(\frac{1}{0}\right)$$

$$\lim_{x\to 0+} \frac{x^2+1}{x} = +\infty$$

$$\lim_{x\to 0-} \frac{x^2+1}{x} = -\infty$$

VA: $x = 0$

HA: $\emptyset$

SA: $\lim_{x\to\infty} \frac{f(x)}{x}$

$$\lim_{x\to\infty} \frac{\frac{x^2+1}{x}}{x} \iff \frac{x^2+1}{x} \cdot \frac{1}{x} = \frac{x^2+1}{x^2}$$

$$\lim_{x\to\infty} \frac{x^2+1}{x^2} = \lim_{x\to\infty} \frac{x^2}{x^2} = 1$$

SA: $y = x$

$$\lim_{x\to\infty} (f(x) - x)$$

$$\lim_{x\to\infty} \frac{1}{x} = 0$$

$y = x$

$$f(x) - A = x + \frac{1}{x} - x = \frac{1}{x}$$

x		0	
$f(x) - A$	$-$	|	$+$

$x \to -\infty \implies f(x)$ onder A

$x \to +\infty \implies f(x)$ boven A

References